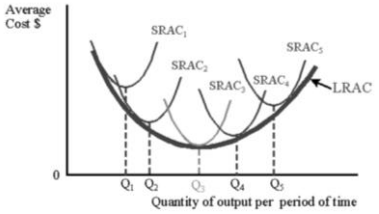
SYLLABUS - LOGISTICS

Economics 20100

The Elements of Economic Analysis II

Winter Quarter 2024



Economies of Scale Image by James R Martin

Prerequisites

This class will assume you are comfortable with Econ 20000 and its prerequisites.

Measured Learning Objectives

Identify economic concepts of the firm and general equilibrium (GE).

Construct a firm-choice problem or a GE problem from a provided scenario.

Translate a provided firm or GE problem into different analysis settings.

Solve a firm or GE problem under different specifications or scenarios.

Synthesize solutions with economic knowledge to extrapolate the implications for agents in the economy.

Description

Department Website: This course is a continuation of ECON 20000. The first part of this course discusses markets with one or a few suppliers. The second part focuses on demand and supply for factors of production and the distribution of income in the economy. This course also includes some elementary general equilibrium theory and welfare economics.

Explained: In Econ 20000 you studied consumer choice – you examined how to mathematically model people's choices. In this class, we begin by thinking about producer choice – how do firms make choices about how much to produce? How does the market structure (the number of firms and consumers) impact that decision? What are the costs and benefits to an economy from having a monopoly, or a duopoly, or competitive market through the lens of economic analysis?

Textbooks

I will post references to chapters by Banerjee, Lima, and Varian in the Topic Summaries to augment or provide details for topics from class. I do not assign mandatory homework or reading from any textbook, though there may be mandatory topical readings assigned in the homework. Those, however, will be provided on Canvas and do not need to be purchased.

	(L) Lima, Victor. Lecture Notes for Econ 20100 Honors. Available for purchase from the Social Sciences Copy Room (SS 103). It is generally at a level that is too high for this course.
	(V) Varian, Hal. Intermediate Microeconomics with Calculus. (There is a non-calculus version available, but don't get that one). A classic that is too low-level in some chapters and can skip some steps.
	(B) Banerjee, Samiran. Intermediate Microeconomics: A Tool-Building Approach. A newer book that is better than Varian at emphasizing the underlying math, but can omit some nuance that Varian discusses.
	(TR) Topical Readings. Articles or news stories that integrate the course content with current events or context. Will be distributed on Canvas or referenced in homework if a required reading.

Econ 20100 SCHEDULE

	CLASS / *Discussion*	Office Hours
MO	(1) 1:30-2:50pm (SHFE 021) (2) 4:30-5:50pm (Rosenwald 011)	9:30am-11:30am SHFE 428
WE	(1) 1:30-2:50pm (SHFE 021) (2) 4:30-5:50pm (Rosenwald 011)	9:00am-11:00am (Advising Only) SHFE 428
TH	-	3:00pm-5:00pm SHFE 428
FR	(1) *1:30-2:20pm (Harper 130)* (2) *1:30-2:20pm (Stuart 104)*	-
		Dr. G.C. Pieters
preferred salutation		Dr./Prof. Pieters she/her

STRUCTURE

In most classes, I will show relevant theory with an application to a real-world situation. Each class meeting introduces a "Quick Response Exercise" (QRE): an economic question to be discussed in groups with analysis ultimately shared with the entire class. Discussion sections revisit mathematical techniques and problems from class and introduce or revisit sample problems closely related to the homework. Discussion section material is selected and designed by the TA.

Homework questions are not a regurgitation of in-class examples. I design the homework assignment to evaluate whether you comprehend the material sufficiently to: (1) understand the question, and (2) complete the economic analysis of that question. The expectation is the same for quiz questions as you must understand the material, not simply memorize definitions or techniques. In addition to class attendance, I assume you will work on the provided practice problems, attend the discussion sections, and visit class hours or post on the Discussion board as needed. In contrast to the class-discussion based QRE and group-based homework, the biweekly quizzes are an individual-level assessment of your knowledge and understanding of the course material.

GRADES CALCULATION

Aggregate Individual Score, $\boldsymbol{AIS_i}$

Your *aggregated individual score* (AIS_i) will not be rounded and is calculated from the combination of the three graded components of this class: Homework, Quick Response Exercises (QRE), and Quizzes.

- Formative Assessment Component (30 points) = Homework Points + QRE Points + Bug Bounty Points
  - The formative assessment component of your grade cannot exceed 30 points. It is the sum of all your homework, QRE, and bug bounty points.
- Test Component (120 points): The sum of your 4 highest Quiz scores.
  - Each quiz is worth 30 points.

$$AIS_i\ (150) = Formative\ Component\ (30) + Quiz\ Component(120)$$

	D	D+	C-	C	C+	B-	B	B+	A-	A
Default										
Fixed (% out of 150)	90 60%	99 66%	102 68%	105 70%	114 76%	117 78%	120 80%	129 86%	132 88%	135 90%
Course Curves (cutoff used if lower than "Default")										
Curve 1: Distribution $\mu = Average\ AIS_i$ $\sigma = Std.\ Dev.$	$\mu - \frac{3}{2}\sigma$	$\mu - \frac{5}{4}\sigma$	$\mu - \sigma$	$\mu - \frac{3}{4}\sigma$	$\mu - \frac{1}{4}\sigma$	μ	$\mu + \frac{1}{4}\sigma$	$\mu + \frac{3}{4}\sigma$	$\mu + \sigma$	$\mu + \frac{5}{4}\sigma$
Curve 2: Share of Grades			C-: AIS_i of the top 75th			B-: Median AIS_i (50th percentile)			A-: AIS_i of the top 25th	

Grade Cutoffs

You will receive the highest letter grade for which your *aggregated individual score*, AIS_i, exceeds the grade cutoff score: $\boldsymbol{AIS_i \geq C_g}$. If the entire class demonstrates mastery of the material (all scores are higher than the cutoff for an A grade), then the entire class will get an A: there is no grade rationing. You are not in a grade competition with other students in this class. I will not certify your understanding as mastery (an A grade) if it is not deserved.

The highest AIS_i needed to **guarantee** a grade is given by the "Fixed" cutoff (If you have an 80%, you will receive no less than a B). I "curve the class" using one of the two curved cutoffs only if the cutoff is lower than the fixed cutoff. I *will never increase the cutoff above the fixed value*.

A "Pass" for a Pass/Fail grade is a D. You must earn a quality (letter) grade of a C- or higher for the course to count towards any track of the Economics major.

Other Grading Notes

• All your grades and graded work will be available on Gradescope. I will not post scores to Canvas because it cannot calculate grades correctly.

• The class average on assessments may vary greatly from assignment to assignment. Individual assessments will not be curved.

•No "Individual-Only" Extra Credit or "Grade Nudges". There will be NO additional grade assistance beyond the curves above. I ignore (and find insulting) all emails that ask for bonus assignments or a "grade nudge". **Everyone has equal opportunity to earn an A grade or to avoid an F**. Grades will not be adjusted to reflect the hours you report working on the course material, your effort, your class participation, your office hours or discussion section attendance, your financial aid status, etc. Your grades reflect only your understanding of the material as evaluated by assessments. Some students may be able to study for 10 minutes and obtain an A, while others can spend an entire week studying and receive an F.

CALENDAR

The following is an outline of all due dates. As the course progresses, the Lecture Topics associated with each class will be updated to reflect what was covered. If the material changes, the date of assessments will **NOT** change, rather the content will be altered.

Week	Date	Summary[§]	Lecture Topic	Due
1	W, Jan 3	1	Production Functions and Costs [Video]	-
	F, Jan 5		*Discussion: Lagrangian/FOC Math Review*	-
2	M, Jan 8	2	Cost Minimization	QRE 1
	W, Jan 10	3	Profit Maximization	QRE 2
	F, Jan 12		*Discussion: Homework 1*	Homework 1
3	M, Jan 15		*MLK Day – no class*	
	W, Jan 17	4	Firm Supply	QRE 3
	F, Jan 19		*Discussion: Quiz Administered*	**Quiz 1**
4	M, Jan 22	5	Partial Equilibrium	QRE 4
	W, Jan 24	6	Market Interventions	QRE 5
	F, Jan 26		*Discussion: Homework 2*	Homework 2
5	M, Jan 29	7	Industry Supply	QRE 6
	W, Jan 31	8	General Equilibrium Production Economy	QRE 7
	F, Feb 2		*Discussion: Quiz Administered*	**Back-up Quiz A**
6	M, Feb 5	9	Welfare Theorems and the Social Planner Problem	QRE 8
	W, Feb 7	10	Externalities	QRE 9
	F, Feb 9		*Discussion: Homework 3*	Homework 3
7	M, Feb 12	11	Monopoly	QRE 10[‡]
	W, Feb 14	12	Game Theory	QRE 11[‡]
	F, Feb 16		*Discussion: Quiz Administered*	**Quiz 2**
8	M, Feb 19	13	Simultaneous Oligopoly (Bertrand and Cournot Models)	QRE 12[‡]
	W, Feb 21	14	Sequential Games and Stackleberg Oligopoly	QRE 13[‡]
	F, Feb 23		*Discussion: Homework 4*	Homework 4[‡]
9	M, Feb 26*	15	Public Goods and Asymmetric Information	QRE 14[‡]
	W, Feb 28	*16*	*TBD*	QRE 15[‡]
	F, Mar 1		*Discussion: Quiz Administered*	**Back-up Quiz B**
Finals**	TBA		50 minutes, 20 minute break, 50 minutes	Quiz 3[‡], Quiz 4[‡]

[§] Corresponding textbook readings and supplementary readings should be determined from the topic summary posted on Canvas. You should also review the material posted to Canvas (the class slides and practice problems).

*This is the last day to Withdraw.

Pass/Fail must be declared by email **no later than the start of the final exam time slot time.

‡ These assessments may be optional for you: see "Disruption Accommodation" on the Policies Syllabus.

POLICIES AND RESOURCES

Documentation posted on Canvas: Econ 20100 Syllabus Policies - Winter 2024 & Econ 20100 Syllabus Resources - Winter 2024.